SDG Lens: Teammate Brief

What it does, why it makes sense, and what results we got

TL;DR

SDG Lens is a prototype that reads an SDG-related text chunk, predicts one or more of the 17 Sustainable Development Goals, and shows the tokens the model attended to most. It is useful because it turns a black-box SDG label into a prediction plus inspectable evidence.

The Intuition

- **Normal SDG classifier:** "this text is SDG 11."
- **SDG Lens:** "this text is SDG 11, and the model focused on words like *housing*, *urban*, and *cities*."
- The explanation is not a human-validated rationale; it is an **attention-based proxy** for what the model emphasized.

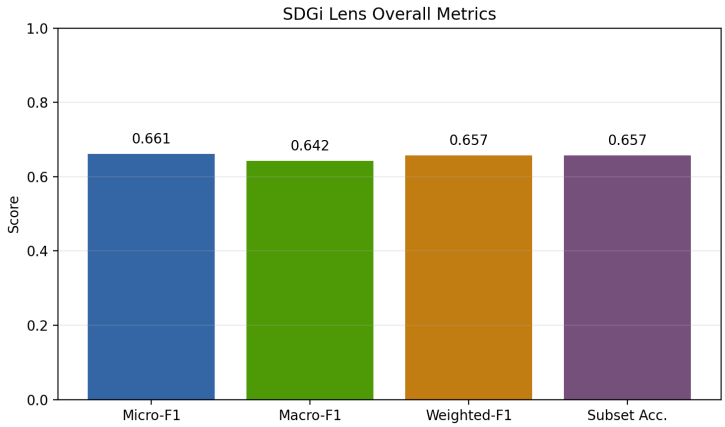
How It Works

`text chunk` → `MiniLM/BERT encoder` → `17 SDG scores` → `predicted SDGs + top attended tokens`

- Dataset: SDGi Corpus, English subset for this prototype run.
- Model: compact BERT-family encoder with a 17-label classifier head.
- Training: 2000 train examples, 300 test examples, 3 epochs.
- Fine-tuning: last 2 transformer layers unfrozen; rest mostly frozen.
- Threshold: 0.3, because multi-label tasks often need a lower cutoff.

Main Results

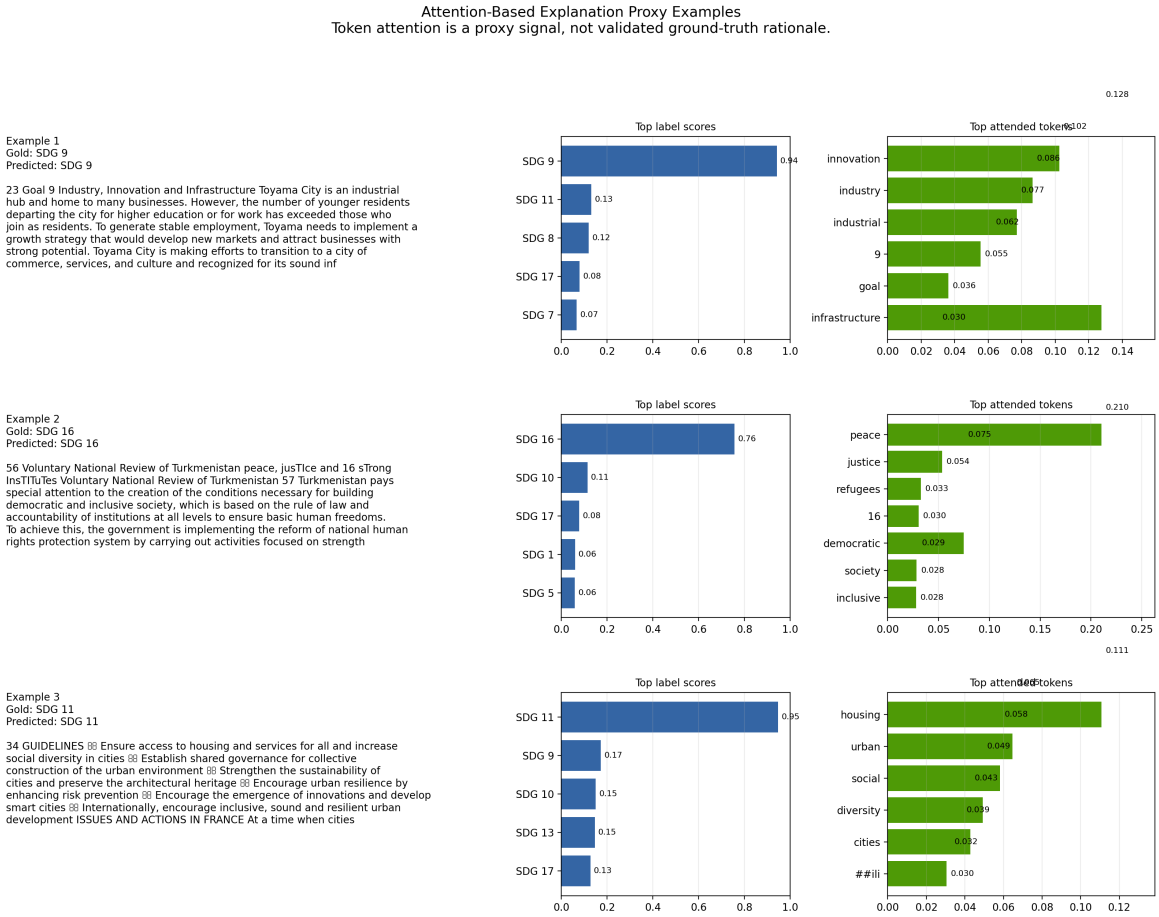
Metric	Score
Micro-F1	0.6615
Macro-F1	0.6425
Weighted-F1	0.6575
Subset accuracy	0.6567
Avg. true labels	1.63
Avg. predicted labels	1.36



Plain-English reading: the model is performing reasonably for a prototype. It is not just producing labels; it also gives a lightweight way to inspect whether the prediction is based on sensible SDG-related terms.

Example Explanations

Each row below shows the text, gold/predicted SDGs, top label scores, and top attended tokens. This is the most useful figure for explaining the idea to someone quickly.



What We Can Say

- The prototype demonstrates that explainable SDG classification is feasible.
- Attention tokens often align with meaningful SDG concepts such as *infrastructure*, *cities*, *housing*, and *digital*.
- The output is easy to report: predicted labels, scores, and token evidence.

What We Should Not Overclaim

- Attention is not causal proof and not a validated human rationale.
- The evaluation is prototype-scale, using an English subset.
- The threshold was fixed, not tuned on a separate validation set.
- SDGi has document-level labels only, so explanation quality cannot be directly scored.

Bottom Line

SDG Lens is a compact proof of concept: it classifies SDG text and provides enough token-level inspection to make the result understandable to a human reader. For reporting, the strongest story is: **we moved from black-box SDG labels to SDG labels with attention-based evidence.**